Hyunsik Jeon

Designing End-to-End AI Systems — Portfolio

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FEATURED PROJECT 01

IAMX6 — Naver Blog Automation Desktop App Hyunsik Jeon (Solo · Private Repo)

Electron · Puppeteer-core · Node.js · AI/LLM · electron-builder (Windows portable .exe / macOS dmg)

Key takeaway: a Naver blog growth tool shipped as an Electron desktop app (not a web app). Puppeteer drives the real browser, an AI model writes context-aware comments, and mutual-neighbor requests, comment/like automation, visitor auto-reply, and blog analytics all ship as one portable .exe.

Technical Pipeline Detail

1. **Desktop architecture** Electron main/preload/renderer cleanly separates UI from automation backend.
2. **Browser automation** Puppeteer-core drives a real Naver session (mutual-neighbor, commenting, likes).
3. **AI comment generation** AI writes context-aware comments with tone presets (friendly / polite / short).
4. **Blog analytics** Blog DNA diagnosis, keyword analysis, and post-level analytics modules.
5. **Distribution** electron-builder packages a single Windows portable .exe plus macOS dmg.

FEATURED PROJECT 02

KOKKOK Garden — K-Beauty Multi-region E-commerce

Hyunsik Jeon (Solo)

Next.js 16 · TypeScript · Supabase · Tailwind CSS 4 · OpenAI GPT-4o mini · Vercel

Live: https://kokv2.vercel.app

Key takeaway: stood up a full stack — frontend → backend → DB → chatbot — in days via vibe coding. Korean users are routed to a purchasable `/kr` store; international users hit a viewing + AI-chatbot `/gl` store. GPT-4o mini auto-translates product data into 5 languages.

> **Technical Pipeline Detail**
>
> **1. Multi-region routing** KR/GL split via `x-user-country` header, six language routes.
>
> **2. AI translation** GPT-4o mini translates products into 5 languages (EN/CN/JP/VN/TH), cached 24h via `unstable_cache`.
>
> **3. AI chatbot** K-beauty consultation chatbot embedded in the GL store.
>
> **4. Unified backend** Supabase (PostgreSQL + Auth + Storage) handles admin auth, product CRUD, image upload.
>
> **5. Deploy** Vercel (https://kokv2.vercel.app).

FEATURED PROJECT 03

AIIM — AI Influencer Matching

Hyunsik Jeon (Solo)

Next.js 16 · TypeScript · Playwright · LLM SDK · Reddit API (snoowrap) · Tailwind CSS 4

Key takeaway: replaces gut-feeling influencer matching with a scraping + LLM-reasoning system. Playwright plus Reddit's public API gather influencer and community signal, and an LLM scores brand-fit with rationale.

> **Technical Pipeline Detail**
>
> **1. Data collection** Playwright dynamic scraping + Reddit public API (snoowrap) for influencer/community signal.
>
> **2. AI matching** LLM-based reasoning over brand traits vs. influencer content, outputting fit score + rationale.
>
> **3. Outreach log** `dm-log` module tracks DM history and status.
>
> **4. End-to-end loop** Collection → normalization → LLM analysis → recommendation inside a single Next.js app.

FEATURED PROJECT 04

Top Award — IBM & RedHat AI Transformation Program

Vaccine Daily (AI News Pipeline)

Hyunsik Jeon and 2 others

React · FastAPI · PostgreSQL · ChromaDB · GPT-4o-mini · ko-sroberta · HDBSCAN · GraphRAG · Docker

An automated news intelligence pipeline that crawls 10 major media outlets every 5 minutes and uses a multi-agent workflow to generate verified, hallucination-free personalized reports.

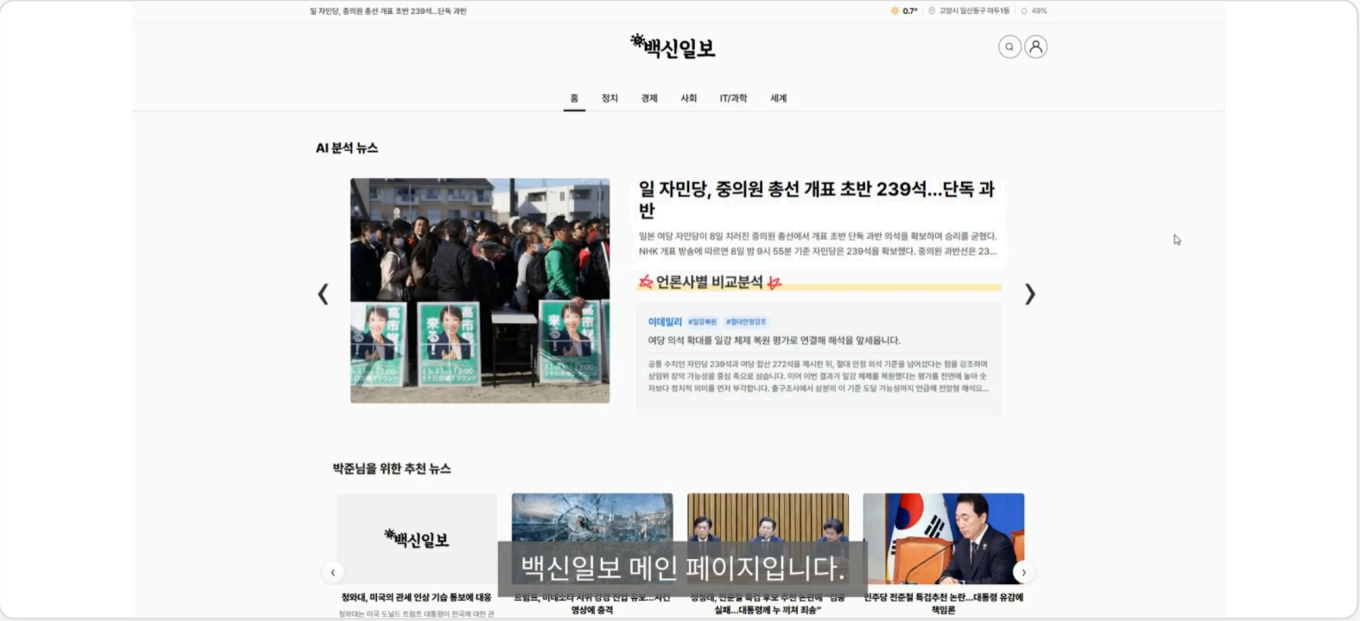
Technical Pipeline Detail

1. Crawling Real-time automated data collection from 10 media outlets via Selenium & BS4.

2. Clustering Token-efficient group analysis of similar news using HDBSCAN and ko-sroberta.

3. AI News Article Generator Multi-Agent AI (Writer → Critic → Editor) for factual verification and summarization.

4. GraphRAG Entity-Relation extraction and Knowledge Graph construction for Media Tone of Voice analysis.



FEATURED PROJECT 05

EPL Match Prediction AI

Hyunsik Jeon (Personal Use)

Python · TensorFlow · Reinforcement Learning · Monte Carlo · XGBoost · LightGBM · FastAPI · Pandas

A high-precision forecasting system analyzing data from the EPL and other major European leagues. It maximizes predictive accuracy by ensembling deep learning models with reinforcement learning strategies.

6-Step System Architecture

Step 1 Data Scrapping: Automated pipeline for match data and global bookmaker odds collection.

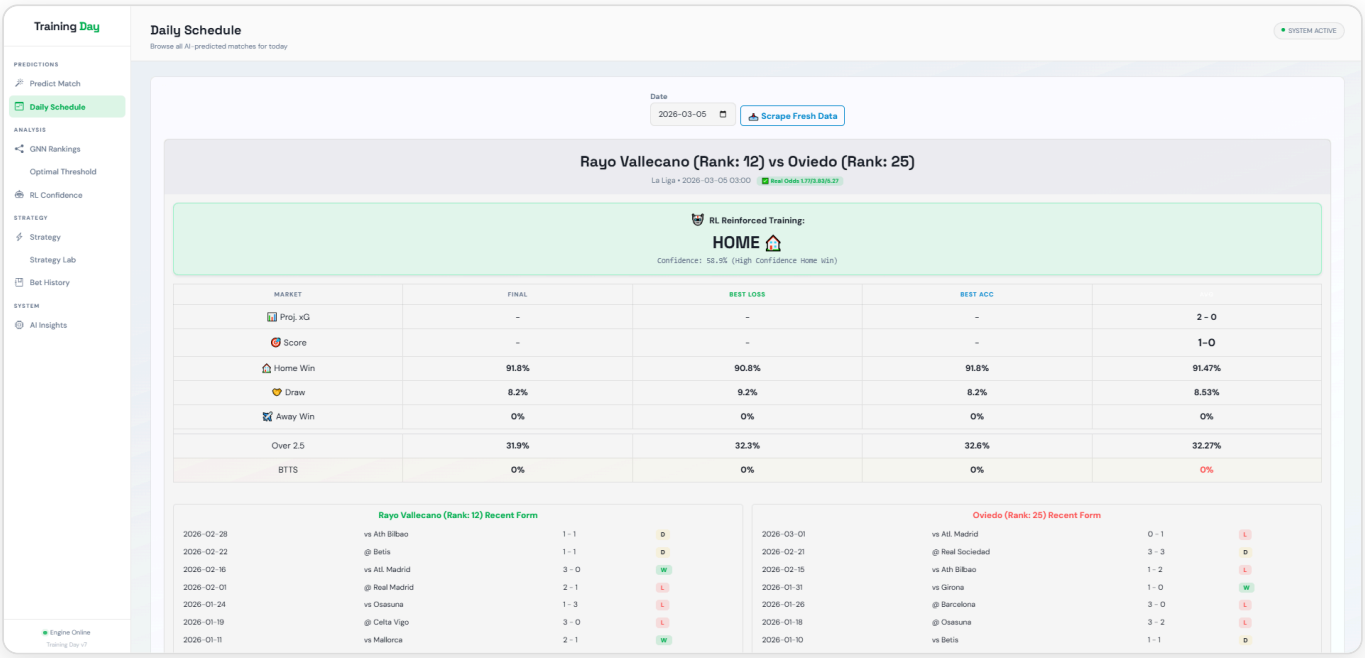
Step 2 Predictive Ensembling: Triple architecture — TensorFlow DL + XGBoost + LightGBM.

Step 3 Confidence Optimization: Reinforcement Learning (RL) agent for tuning logical confidence thresholds.

Step 4 Probabilistic Reasoning: Monte Carlo simulation for calibrated outcome probabilities.

Step 5 Dynamic Tuning: Adaptive sequence length optimization based on league-specific volatility.

Step 6 Power Ranking: ELO-based strength metrics and GNN-based dominance analysis.



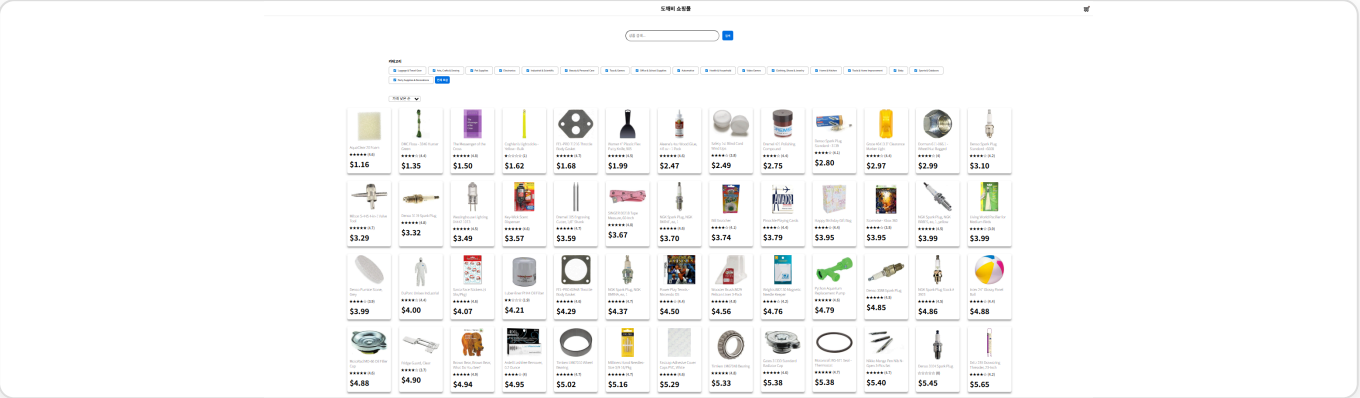
OTHER PROJECTS

Goblin Shopping Mall (E-commerce SPA)

Hyunsik Jeon and 3 others

React · JavaScript · CSS · Vercel

A fully-featured e-commerce platform with 18+ category filters and real-time search. Focused on intuitive UX and high-performance frontend state management.



Crazy Arcade Web Replica

Hyunsik Jeon and 3 others

React · FastAPI · WebSocket · MySQL

A real-time multiplayer game synchronization project using WebSockets. Implemented complex physics and room-state management for smooth gameplay.

My Diary (Emotion Log App)

(Personal Use for Education)

React · Context API · localStorage

A personal journal application focusing on frontend state persistence and custom UI interactions using browser's local storage.